HTTP://WWW.MAKAUT.COM

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : IT-802C

OPTICAL NETWORKING

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for the following : $10 \times 1 = 10$

i) A step index fibre has a core with refractive index 1·57 and a cladding with refractive index 1·41. Its numerical aperture is

a) 0·69 b) 0·75

c) 0·57 d) 0·65.

ii) Which optical source-detector combination is suitable for high bandwidth long haul communication ?

a) LED-pin diode b) LASER diode-APD

c) LED-LASER diode d) LED-APD.

HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM

iii) The suitable material for an optical detector is

a) a direct band gap semiconductor

b) an indirect band gap semiconductor

c) a metal

d) none of these.

iv) For short haul optical communication, the suitable optical fibre is

a) single mode step index fibre

b) multimode step index fibre

c) graded index single mode fibre

d) graded index multimode fibre.

v) Which of the following materials is suitable for making a light source operating in the near infrared region (800-900 nm) ?

a) GaAlAs b) GaAs

c) Si d) None of these.

vi) The transmission mode which uses start and a stop bits in addition with the data bits is

a) Synchronous b) Asynchronous

c) Isochronous d) none of these.

vii) Virtual circuit switching is implemented in

a) Physical layer b) Network layer

c) Data link layer d) Transport layer.

viii) Erbium-doped fibre amplifiers (EDFA) operates at which of the following windows ?

a) Around 1300 nm

b) Around 1550 nm

c) Both the windows

d) None of these.

HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM

ix) Which of the following fibres is suitable for wavelength division multiplexing signals ?

a) Dispersion optimized

b) Dispersion shifted

c) Dispersion flattened

d) Any one of these.

x) The maximum frame size for an FDDI frame is

a) 4500 bytes b) 4000 bytes

c) 2500 bytes d) 1250 bytes.

GROUP – B
(Short Answer Type Questions)

Answer any *three* of the following. 3 × 5 = 15

2. What are acceptance angle of and normalized frequency of an optical fibre ? Write the value of normalized frequency of a single mode step index fibre. How can the number of guided modes through the core of a graded index fibre be determined from the same of it ? 3 + 1 + 1

3. Write the features of circuit switching. What is virtual circuit identifier ? 4 + 1

4. Discuss on burst switching.

5. Explain the DQDB architecture with diagram. Describe its working principle. 3 + 2

6. Mention the advantages and disadvantage of SDMA.

HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM

GROUP – C
(Long Answer Type Questions)

Answer any *three* of the following. 3 × 15 = 45

7. a) Explain the functions performed in the data link layer of a packet switched network.

b) What is service point addressing ? Explain.

c) Write the advantages and disadvantages of a packet switched network over a circuit switched network. 7 + 3 + 5

8. a) Explain the working of an Erbium-doped fibre amplifier (EDFA).

b) Write the advantages of a fibre Bragg grating.

c) How does a first generation tunable wavelength converter work ? Explain from its block diagram. 6 + 4 + 5

9. a) What are common channel signalling (CCS), associated and non-associated signalling in telecom networks ?

b) Write the features of ALOHA and CSMA/CD protocols in a single channel broadcast network.

c) What are the advantages and problems in all-optical packet switching ? 4 + 6 + 5

10. a) What are the main features of SONET/SDH ring ?

b) Describe different layers of Fibre Channel network.

c) Mention the working principle of OCDMA. 5 + 5 + 5

11. Write short notes on any *three* of the following : 3 × 5

a) Tunable transmitter-Fixed receiver

b) Reserved access

c) Packet buffering

d) High Performance Parallel Interface (HIPPI)

e) Shuffle net.

HTTP://WWW.MAKAUT.COM